

Overview

Part 05 —  Generative AI · 05-genai/README.md

Module 05 — Generative AI

Where you are: Stop 5 of 16. **Time:** ~20–25 hours · **Prereq:** Module 02 [02-deep-learning] (VAEs, backprop), Module 03 [03-nlp] (attention/cross-attention).

Modules 03–04 taught you a generative model without calling it one — a language model samples from $P(\text{next token} \mid \text{context})$. This module widens the lens to pixels, video, and audio, and spends most of its time on the family that now dominates all three: **diffusion**. You will train a real (tiny) diffusion model from scratch, measure the DDPM-vs-DDIM speed/quality trade-off in numbers, and see classifier-free guidance push fidelity against diversity on a toy dataset — not on faith, on printed output. The second half covers what you actually ship with this: text-to-image pipelines, video and audio generation, vision-language models, and the honest (unsatisfying) state of evaluating any of it.

Learning objectives

By the end of this module you can:

1. Explain why diffusion displaced GANs/VAEs as the default generative architecture, and name each family's fatal-ish weakness.
2. Derive the forward (noising) and reverse (denoising) diffusion processes, and implement DDPM training and sampling from scratch in NumPy.
3. Explain DDIM sampling and the accuracy/speed trade-off of fewer reverse steps, with numbers, not vibes.
4. Implement classifier-free guidance and describe the fidelity-vs-diversity trade-off as the guidance scale increases — including where it breaks down.
5. Draw the U-Net-to-DiT architecture shift and connect it to the same scaling story that took over NLP.
6. Trace a text-to-image pipeline end to end: text encoder, cross-attention, scheduler, VAE decode, and where ControlNet/img2img/inpainting slot in.
7. Name generative AI's real failure modes and unresolved questions — hallucinated detail, deepfake misuse, watermarking limits, training-data licensing — and say which module owns each one.

Suggested path

#	Do this	File	Time
1	Diffusion mechanism, GAN/VAE comparison, DiT	notes/01-core-concepts.md	3h
2	Pipelines, multimodality, evaluation, failure modes	notes/02-multimodal-and-practice.md	3h
3	Train + sample a DDPM from scratch	code/diffusion_from_scratch.py	2h
4	Measure DDPM vs. DDIM and the guidance-scale sweep	code/guidance_and_sampling.py	2h
5	Present it back	slides/	1h
6	Do the exercises	lab/EXERCISES.md	8h
7	Read the papers	papers/PAPERS.md	5h

The 14 terms you must own

diffusion · forward/reverse process · noise schedule · DDPM · DDIM · classifier-free guidance · latent diffusion · U-Net · DiT (diffusion transformer) · cross-attention · ControlNet · mode collapse · FID · CLIP score

Where this connects

Topic here	Where it returns
VAEs, latent bottlenecks	../02-deep-learning/ — latent diffusion's direct ancestor
Cross-attention, transformer blocks	../03-nlp/ — the same mechanism doing double duty; DiT is the same architecture shift NLP already went through
Scaling laws, multimodal/vision-language models	../04-llm/ — LLaVA-style splicing of image tokens into an LLM
Deepfakes, voice cloning, watermark limits	../13-ai-security/ — social-engineering and synthetic-media threat models
Training-data licensing, output copyright	../14-ai-compliance/ — audit trails for model/data provenance
FID/CLIP score's limits as a metric	../15-ai-evals/ — the image-generation flavor of "no single metric captures 'good'"

Exit check

You can run `code/diffusion_from_scratch.py` and `code/guidance_and_sampling.py` unmodified, explain every printed number (forward-process std, denoiser MSE floor, DDIM steps-vs-quality, guidance fidelity/diversity), and then — without looking at the code — sketch the DDPM training loop and the classifier-free-guidance sampling equation on a whiteboard.